

AI SOFTWARE DEVELOPMENT INTERNSHIP

EMAN MASOOD (BSSE)

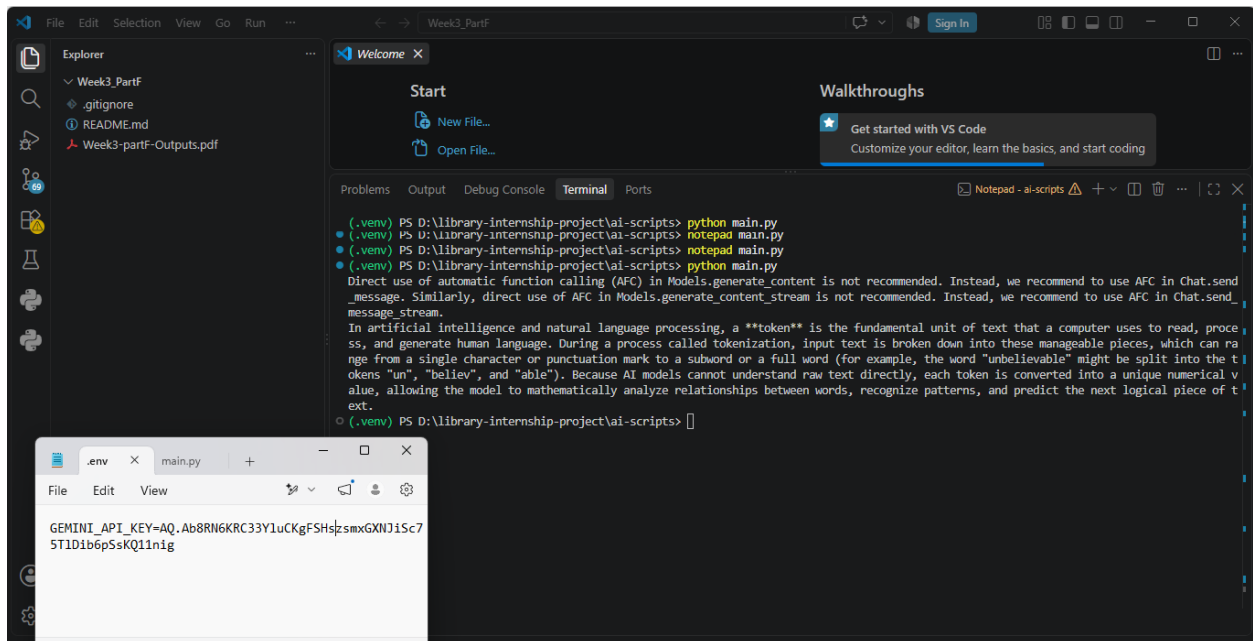
Week 3: SQL Server & Entity Framework Core • API/Angular
Integration Git & GitHub Workflow • AI & Python Foundations

PART F — GIT & GITHUB WORKFLOW (NEW THIS WEEK)

Practice Tasks

1. Run the script above and confirm you get a real response back.

Output:



The screenshot shows a Windows environment with Visual Studio Code (VS Code) and a Notepad window. In VS Code, the Explorer sidebar on the left shows a project named 'Week3_PartF' containing files '.gitignore', 'README.md', and 'Week3-partF-Outputs.pdf'. The main editor area displays the 'Terminal' tab, which shows a series of commands and their outputs in a PowerShell prompt. The commands executed are:

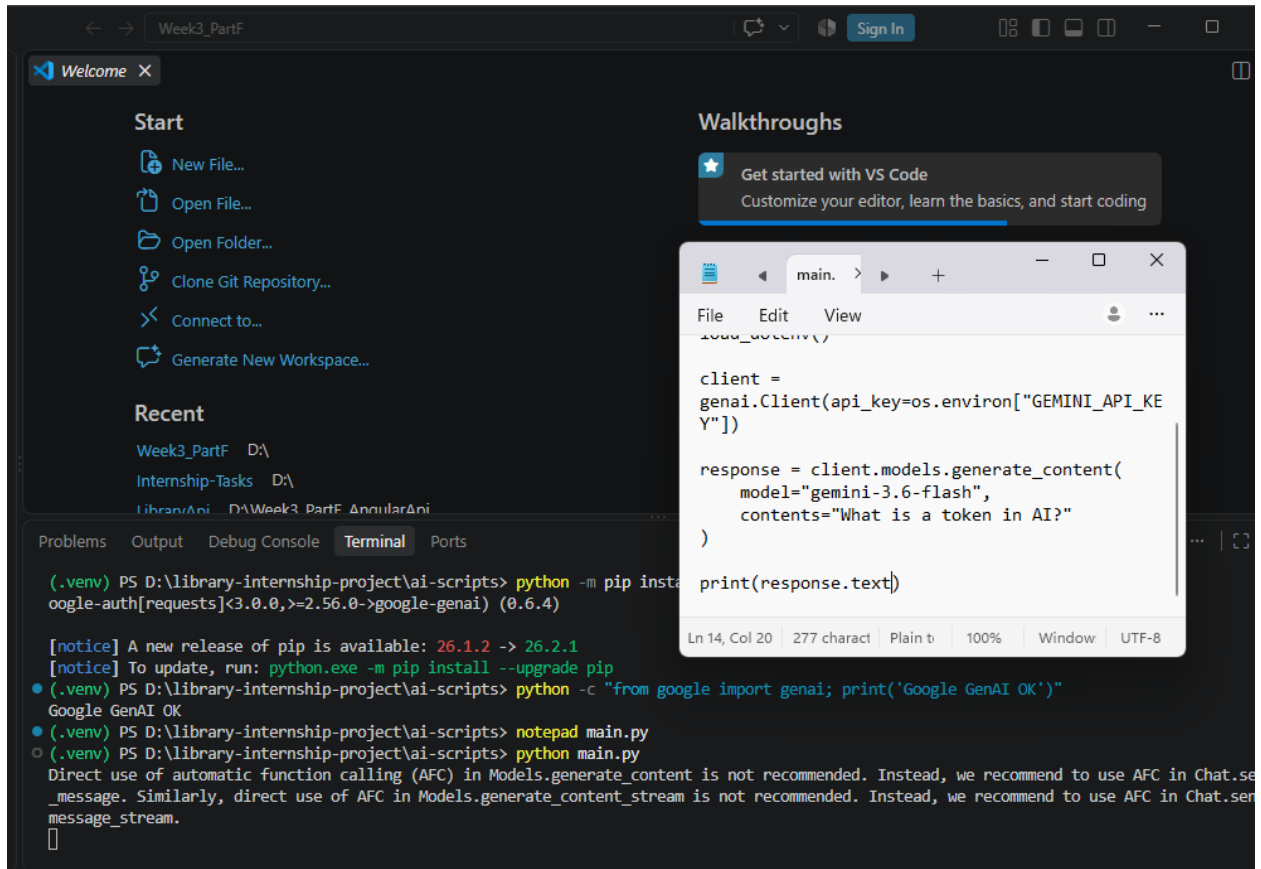
```
(.venv) PS D:\library-internship-project\ai-scripts> python main.py
(.venv) PS D:\library-internship-project\ai-scripts> notepad main.py
(.venv) PS D:\library-internship-project\ai-scripts> notepad main.py
(.venv) PS D:\library-internship-project\ai-scripts> python main.py
```

The output of the last command shows a message about Automatic Function Calling (AFC) and a detailed explanation of tokens in natural language processing. The Notepad window, titled '.env', contains the following text:

```
GEMINI_API_KEY=AQ.Ab8RN6KRC33Y1uCKgF5HszsmxGXNj1Sc7
5T1D1b6pSsKQ11nig
```

2. Change the prompt three different ways — a plain question, a direct instruction, and a role-based prompt ("You are a strict librarian who only answers in one sentence...") — and write down how the output changed for each.

Output:



The screenshot shows the Visual Studio Code editor interface. The left sidebar contains the 'Start' and 'Recent' sections. The main editor area displays a Python script in a file named 'main.py'. The script imports the 'genai' module and uses the 'generate_content' method to generate a response to the prompt 'What is a token in AI?'. The terminal at the bottom shows the execution of the script, including pip installation and execution commands, and the output 'Google GenAI OK'.

```
client =
genai.Client(api_key=os.environ["GEMINI_API_KEY"])

response = client.models.generate_content(
    model="gemini-3.6-flash",
    contents="What is a token in AI?"
)

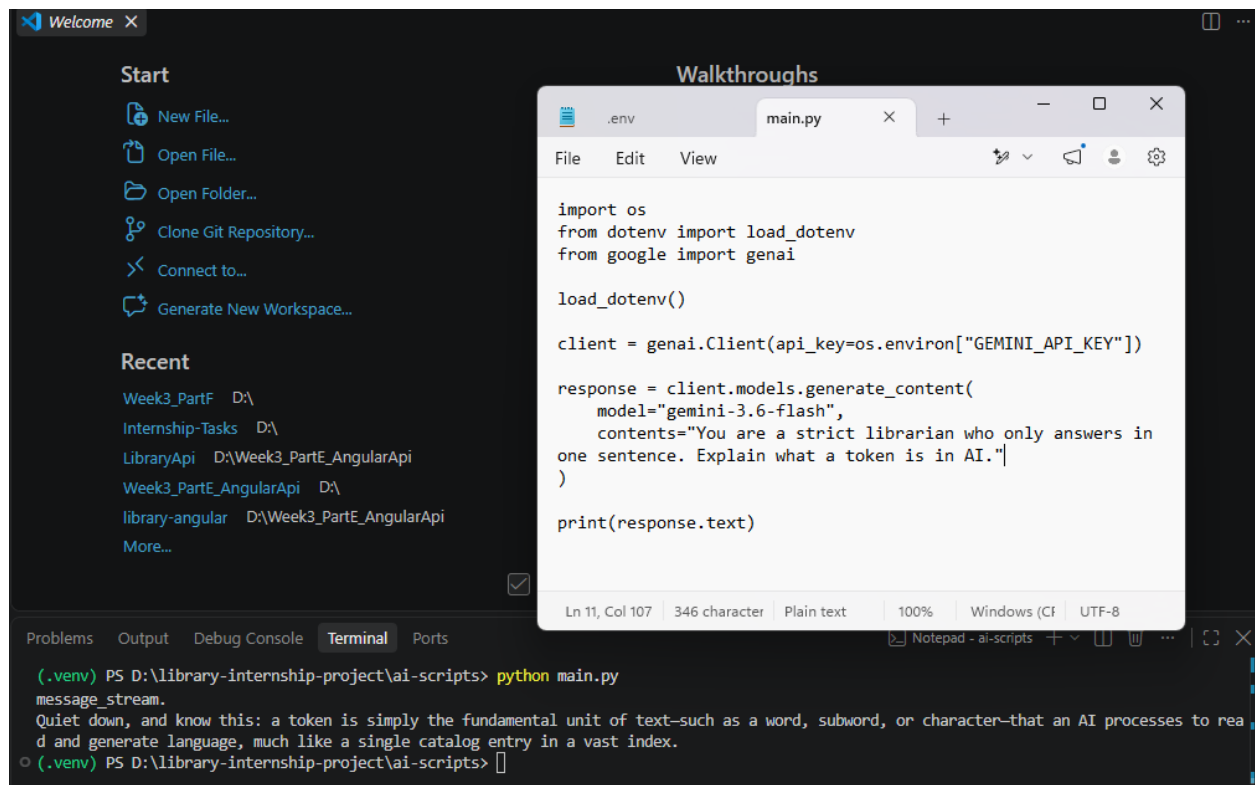
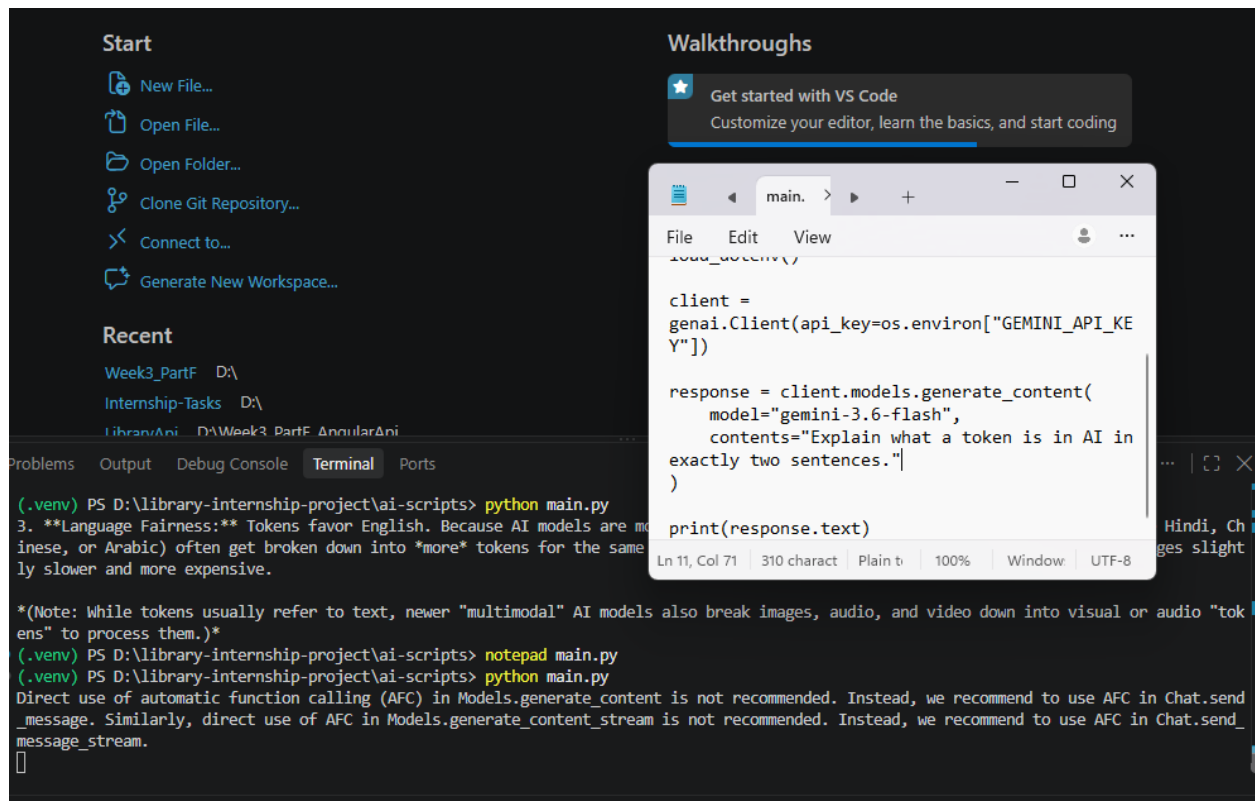
print(response.text)
```

```
(.venv) PS D:\library-internship-project\ai-scripts> python -m pip install google-auth[requests]<3.0.0,>=2.56.0->google-genai (0.6.4)

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\library-internship-project\ai-scripts> python -c "from google import genai; print('Google GenAI OK')"
```

Google GenAI OK

```
(.venv) PS D:\library-internship-project\ai-scripts> notepad main.py
(.venv) PS D:\library-internship-project\ai-scripts> python main.py
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.session.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.session.send_message_stream.
```



Plain question:

The model gave a general explanation.

Direct instruction:

The model followed the instruction to answer in exactly two sentences.

Role-based prompt:

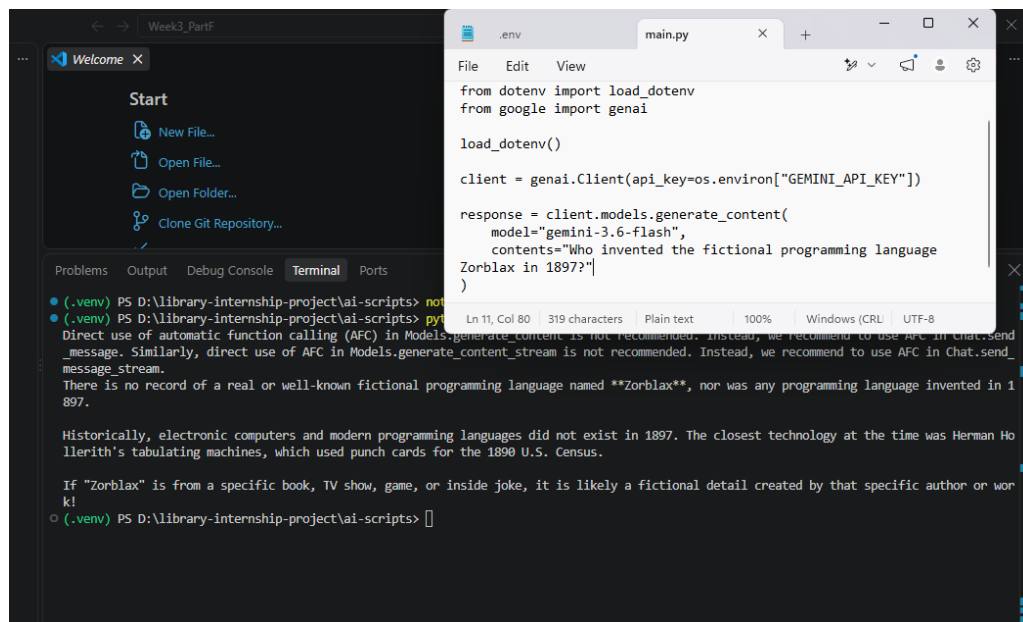
The model followed the librarian role and answered in one sentence.

Overall observation:

Changing the prompt changed the response length, format, and style.

3. Deliberately ask the model something obscure or made-up and observe whether it hallucinates an answer instead of saying it doesn't know.

Output:



The screenshot shows a VS Code editor with a Python file named `main.py` and a terminal window. The Python code imports `load_dotenv` and `genai` from `google`, initializes a client with a Gemini API key, and generates content from the `gemini-3.6-flash` model with the prompt "Who invented the fictional programming language Zorblax in 1897?". The terminal output shows the model's response, which is a detailed but incorrect history of the fictional language Zorblax, claiming it was invented in 1897 by Herman Holterith.

```
.env main.py
File Edit View
from dotenv import load_dotenv
from google import genai

load_dotenv()

client = genai.Client(api_key=os.environ["GEMINI_API_KEY"])

response = client.models.generate_content(
    model="gemini-3.6-flash",
    contents="Who invented the fictional programming language
Zorblax in 1897?"
)

Ln 11, Col 80 | 319 characters | Plain text | 100% | Windows (CRLF) | UTF-8

(.venv) PS D:\library-internship-project\ai-scripts> not
(.venv) PS D:\library-internship-project\ai-scripts> py
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
There is no record of a real or well-known fictional programming language named **Zorblax**, nor was any programming language invented in 1897.

Historically, electronic computers and modern programming languages did not exist in 1897. The closest technology at the time was Herman Holterith's tabulating machines, which used punch cards for the 1890 U.S. Census.

If "Zorblax" is from a specific book, TV show, game, or inside joke, it is likely a fictional detail created by that specific author or work!
(.venv) PS D:\library-internship-project\ai-scripts> |
```

Hallucination Test:

I asked the model about a fictional programming language called "Zorblax" supposedly invented in 1897.

Observation:

The model did not confidently invent a person or historical fact. Instead, it stated that there is no known programming language named Zorblax and explained why the claim was historically unlikely.

Conclusion:

The model was able to recognize that the information was fictional rather than presenting a fabricated answer as fact.

← ↻ <https://github.com/emanmasood0099-em/library-internship-project/pull/1> 🔍 🗺️ ☆ ⚙️ 👤 ... Chat

Conversation 0 Commits 1 Checks 0 Files changed 1

 **emanmasood0099-em** commented 4 minutes ago Owner ...

Updated the README to document the Git workflow used for this project.

The change demonstrates making documentation updates on a feature branch before merging them into main.



 docs: document Git workflow @6ed5b3

 **emanmasood0099-em** merged commit e22afaf into main now Revert

 **Pull request successfully merged and closed**
You're all set — the `feature/git-github-workflow` branch can be safely deleted. Delete branch

 **Add a comment**

Write Preview H B I          

Add your comment here...

☐ Markdown is supported ☐ Paste, drop, or click to add files

Comment

Reviewers
No reviews
Still in progress? [Convert to draft](#)

Assignees
No one—[assign yourself](#)

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

Notifications Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.

1 participant

 **Merged** docs: document Git workflow #1
emanmasood0099-em merged 1 commit into `main` from `feature/git-github-workflow`

The change demonstrates making documentation updates on a feature branch before merging them into main.



 docs: document Git workflow @6ed5b3

 **emanmasood0099-em** merged commit e22afaf into main 1 minute ago Revert

 **emanmasood0099-em** deleted the `feature/git-github-workflow` branch now Restore branch

 **Pull request successfully merged and closed**
You're all set — the branch has been merged.

 **Add a comment**

Write Preview H B I          

Add your comment here...

☐ Markdown is supported ☐ Paste, drop, or click to add files

Comment

Assignees
No one—[assign yourself](#)

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

Notifications Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.

1 participant 

Lock conversation

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).